Limits of AI peer review on real expert review data

Valentin Klotzbücher David Reinstein Lorenzo Pacchiardi
Tianmai Michael Zhang

2026-02-18

Abstract

We benchmark how frontier LLMs critique, rate, and prioritize research, comparing AI-generated evaluations to human expert assessments from The Unjournal. Human ratings serve as a reference signal—the core goal is measuring and diagnosing AI reasoning quality: understanding calibration, failure modes, and systematic "taste" differences between AI and human evaluators.

Introduction

> **Working paper**
>
> This is an evolving working paper. Analysis, metrics, and comparisons are under active development.

Peer review is under strain. Reviewers are hard to find, turnaround times are lengthening, and the system costs an estimated $1.5 billion per year in the United States alone (Aczel, Szaszi, and Holcombe 2021). An influx of AI-generated manuscripts is likely to intensify the bottleneck (Eger et al. 2025). If frontier language models could produce reliable, structured evaluations of research, they might free substantial scientific resources and deliver faster, more detailed feedback.

We test this possibility using The Unjournal's structured human evaluations as a benchmark. We prompt six frontier LLMs—GPT-5 Pro, GPT-5.2 Pro, GPT-4o-mini, Claude Sonnet 4, Claude Opus 4.6, and Gemini 2.0 Flash—with the same rubric and guidelines used by human evaluators, then compare the resulting quantitative ratings and qualitative critiques against human evaluations for 60 economics and social-science working papers. We ask whether frontier AI evaluations can reliably approximate human expert judgment, which models perform best across rating criteria and cost tiers, and whether systematic differences reveal characteristic AI preferences over research.

The Unjournal setting is particularly well suited for this comparison. It commissions paid expert evaluations using a structured rubric covering seven percentile criteria with 90% credible intervals plus journal-tier predictions, and publishes the resulting packages openly—reducing classic gatekeeping motives and increasing reviewer effort. The resulting ratings and critiques still exhibit substantial inter-rater variation; accordingly, we treat human evaluations as a high-quality but noisy *reference signal*, not ground truth. The rich, multi-dimensional data allow us to compare the priorities and calibration of humans and AI models across criteria and domains, while the journal-tier predictions provide an external reference point[1] enabling a human-vs-LLM horse

[1] These represent verifiable publication outcomes, not statements about the "true quality" of the paper.

race. Finally, The Unjournal's pipeline of future evaluations allows for clean out-of-training-data predictions, serving as a live testing lab for prospective validation.

A growing literature examines LLM-assisted peer review, error detection, and AI research evaluation; we survey this work in Section . Our approach differs from prior work by focusing on structured, percentile-based quantitative ratings with credible intervals, comparing against published human evaluations rather than using LLM-as-judge, and curating contamination-aware inputs.

Acknowledgements.

This work is a collaboration with The Unjournal, which has published 55+ detailed evaluation packages building open, quantitative evaluation infrastructure for global-priorities-relevant research in economics, policy, and social science. We thank The Unjournal's evaluation community for generating the structured human assessments that make this benchmarking possible.

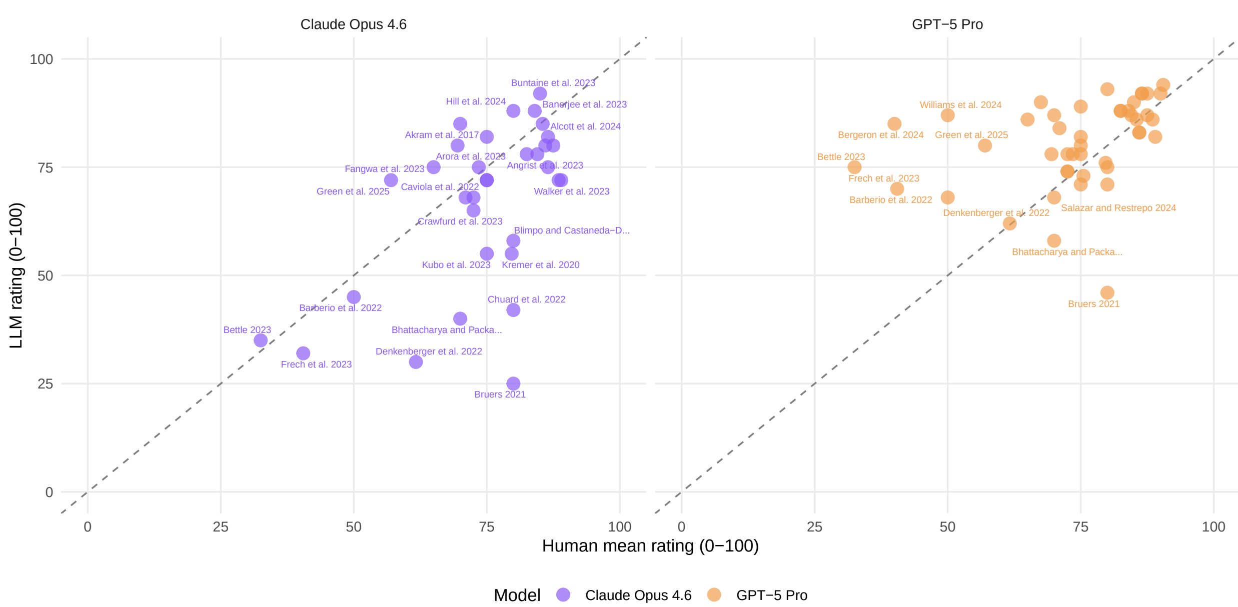
Funding for The Unjournal has been provided by the Survival and Flourishing Fund, the Long Term Future Fund, and EA Funds. Portions of this manuscript were drafted with AI assistance and reviewed by the authors.

Results

We evaluate 6 frontier LLMs against human expert reviews from The Unjournal for 45 papers with both human and LLM evaluations. Each model receives the same PDF, system prompt mirroring The Unjournal rubric, and JSON schema requiring a diagnostic summary plus numeric midpoints and 90% credible intervals for every metric. Full methodological details appear in Appendix: Data and methods.

Overall ratings. Figure 1 compares overall ratings from GPT-5 Pro and Claude Opus 4.6—the two highest-performing reasoning models—against human mean ratings. Points on the diagonal indicate perfect agreement; distance from the line reflects disagreement.

Figure 1: Human vs LLM overall ratings: GPT-5 Pro and Claude Opus 4.6



Both models cluster around the identity line in the 40–80 range but diverge more at the extremes. LLMs tend to compress ratings toward the centre of the scale relative to humans, producing narrower spread—a pattern consistent with alignment training that discourages extreme outputs. Where humans rate a paper very highly or very harshly, the LLM typically pulls toward the middle. Full agreement metrics including Pearson r, Spearman , mean bias, and RMSE for all six models appear in Table 2.

Rank comparison. Figure 2 contrasts the relative *ranking* of papers under human and LLM scoring. Each paper is a curve connecting its human rank (left) to its GPT-5 Pro rank (right). Horizontal lines indicate agreement; orange curves slope upward where the LLM ranks a paper higher than humans did, and green curves slope downward where humans rank it higher.

Many papers sit close to horizontal, indicating broad agreement on relative quality. However, several show pronounced rank shifts—orange curves mark papers the LLM considers top-tier that humans placed lower, while green curves highlight the reverse. These rank swaps suggest characteristic "taste" differences rather than random noise, since they persist across criteria (see the gap heatmap below). The criteria-level Krippendorff's alpha analysis in Table 3 quantifies where human–LLM agreement approaches the ceiling set by human inter-rater variability.

Qualitative critique comparison. Beyond numeric ratings, we compare the substantive critiques each model raises against the consensus issues identified by human experts. The figure below plots coverage (what fraction of human concerns the LLM also identified) against precision (what fraction of LLM-raised issues correspond to human concerns) for each paper, as assessed by GPT-5.2 Pro acting as judge.

Coverage and precision vary substantially across papers. Some papers achieve high scores on both dimensions, indicating strong alignment between human and LLM critiques, while others reveal the LLM missing key human concerns or raising issues absent from the expert consensus. These metrics are themselves LLM-assessed (GPT-5.2 Pro as judge); human validation through manual annotation is underway. Detailed paper-by-paper comparisons with matched issue pairs appear in Appendix: Critiques & Key Issues.

Rating differences by paper. Figure 3 shows Human − LLM differences for each paper and criterion using GPT-5 Pro as the focal model. Green tiles indicate humans rated higher; orange indicates the LLM rated higher. Papers are ordered by their overall rating difference, revealing which papers drive the largest disagreements and whether those disagreements are concentrated on particular criteria or spread across the board.

Rows with uniformly green or orange tiles indicate papers where humans and the LLM disagree systematically across all criteria, not just on one dimension. Columns with consistent colour suggest criteria-level biases. Extended statistical analysis—including Krippendorff's alpha, bootstrap confidence intervals, mutual information, tier correlations, and top disagreement cases—is available in Appendix: Results Ratings.

Discussion

Frontier LLMs achieve moderate-to-strong agreement with human expert evaluators, approaching the ceiling set by inter-rater variability among humans themselves. Correlations with human mean ratings are highest for reasoning-capable models (GPT-5 Pro, Claude Opus 4.6), though even lightweight models (GPT-4o-mini, Gemini 2.0 Flash) perform respectably at a fraction of the cost. Central compression—the tendency for LLMs to pull extreme ratings toward the middle of the

Figure 2: Relative ranking (overall) by GPT-5 Pro and Human evaluators

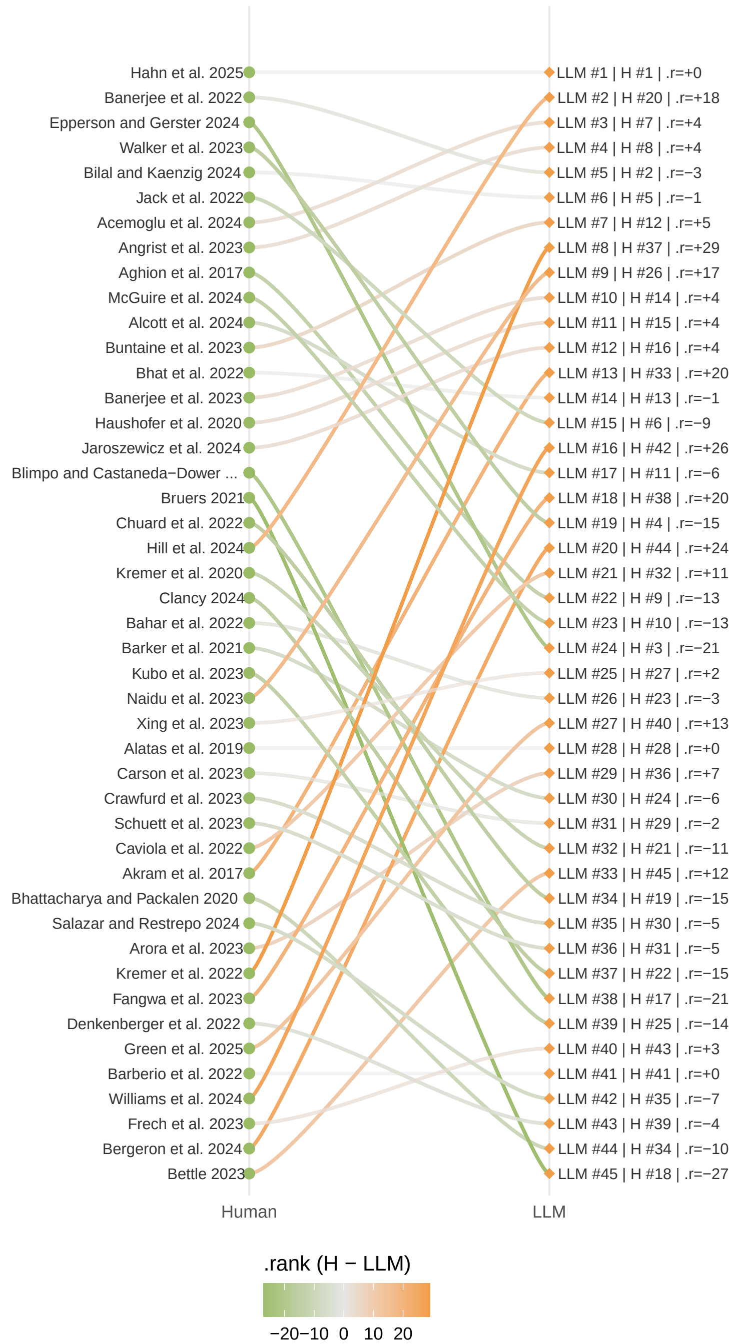
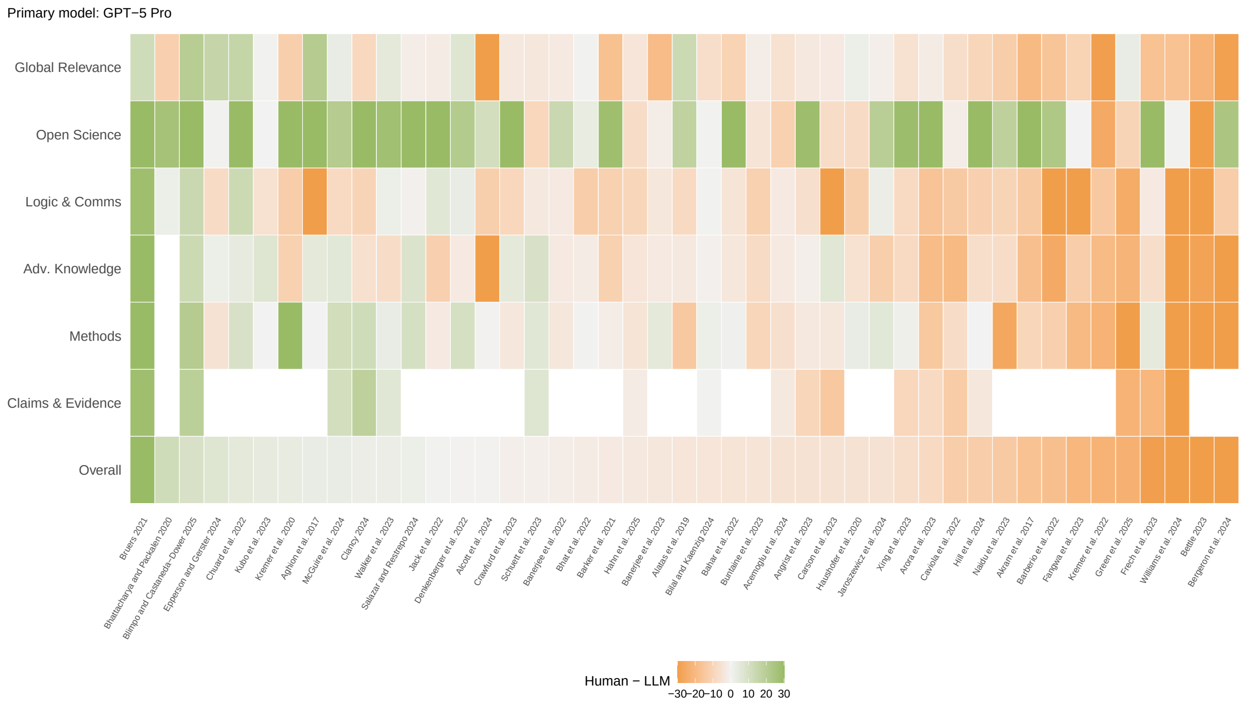


Figure 3: Human − LLM rating differences by paper and criterion (GPT-5 Pro)



scale—is the most consistent pattern across all models, likely reflecting alignment training that discourages confident extreme outputs. Qualitative coverage varies widely across papers: on some, the LLM captures nearly all consensus human concerns; on others, it misses key critiques or raises issues absent from the expert consensus.

Limitations. Several caveats temper these conclusions. Our sample comprises roughly 50 social-science papers specifically selected by The Unjournal for evaluation, not a random draw from the research literature; performance may differ in other fields or on less polished manuscripts. Human evaluations are themselves a noisy reference signal rather than ground truth, with substantial inter-rater variation that caps achievable agreement. We cannot fully rule out knowledge contamination: while we block extrinsic priors in the system prompt, the models' training data may include fragments of these papers or related discussions. Alignment training likely contributes to score inflation and narrower credible intervals than humans provide. All LLM evaluations are single-run; aggregating across multiple runs or temperature settings could change the picture. Finally, the qualitative coverage and precision metrics are themselves LLM-assessed (GPT-5.2 Pro as judge), introducing a further layer of model dependence.

Implications. The cost structure is striking: lightweight models deliver evaluations at under a cent per paper, while reasoning-capable models cost several dollars—still orders of magnitude cheaper than human expert review. This suggests a practical tier structure in which cheap models could serve as rapid screening tools and expensive reasoners could provide deeper assessment where warranted. However, the qualitative gaps we observe—missed critiques, generic issues, and central compression of ratings—argue against full automation of peer review. AI evaluation appears most promising as a supplement: providing fast structured feedback, flagging potential concerns for human reviewers, and enabling systematic comparison across large paper sets that would be infeasible with human effort alone.

Future work. Several extensions follow naturally from this analysis. Content-swap bias tests following Pataranutaporn et al. (2025) would reveal whether LLMs show systematic biases based on author names or institutional affiliations. A journal-outcome prediction horse-race would compare human and LLM tier predictions against actual publication venues. Systematic prompt and model comparisons, including agent-based approaches, could identify configurations that reduce central compression or improve qualitative coverage. Human enumerator validation—employing trained raters to independently assess whether LLMs identify the same consensus critiques—would ground the currently LLM-assessed coverage metrics. Out-of-time validation using papers entering The Unjournal's pipeline would eliminate contamination concerns entirely. Finally, hybrid human-AI evaluation trials would test whether collaboration improves evaluation quality beyond either alone.

References

```
WARNING: One or more problems were discovered while enumerating dependencies.

# /Users/valik/repos/llm-uj-research-eval/results_ratings.qmd ----------------
Error: <text>:9:1: unexpected '/'
8: scales::dollar(total_cost, accuracy = 0.01)
9: /
   ^

# /Users/valik/repos/llm-uj-research-eval/slides_vk/index.qmd ----------------
Error: Parser error: while parsing a block mapping at line 8, column 7 did not find expected ke

Please see `?renv::dependencies` for more information.
```

We used R v. 4.5.2 (R Core Team 2025) and the following R packages: DT v. 0.34.0 (Xie et al. 2025), ggalluvial v. 0.12.5 (Brunson 2020; Brunson and Read 2023), ggbreak v. 0.1.6 (Shuangbin Xu et al. 2021), ggforce v. 0.5.0 (Pedersen 2025a), ggrepel v. 0.9.6 (Slowikowski 2024), glue v. 1.8.0 (Hester and Bryan 2024), here v. 1.0.2 (Müller 2025), irr v. 0.84.1 (Gamer, Lemon, and <puspendra.pusp22@gmail.com> 2019), janitor v. 2.2.1 (Firke 2024), kableExtra v. 1.4.0 (Zhu 2024), knitr v. 1.51 (Xie 2014, 2015, 2025), patchwork v. 1.3.2 (Pedersen 2025b), plotly v. 4.11.0 (Sievert 2020), renv v. 1.0.0 (Ushey and Wickham 2023), reticulate v. 1.44.1 (Ushey, Allaire, and Tang 2025), rmarkdown v. 2.30 (Xie, Allaire, and Grolemund 2018; Xie, Dervieux, and Riederer 2020; Allaire et al. 2025), scales v. 1.4.0 (Wickham, Pedersen, and Seidel 2025), stringi v. 1.8.7 (Gagolewski 2022), tidyverse v. 2.0.0 (Wickham et al. 2019), viridis v. 0.6.5 (Garnier et al. 2024).

Aczel, Balazs, Barnabas Szaszi, and Alex O Holcombe, "A billion-dollar donation: Estimating the cost of researchers' time spent on peer review," *Research integrity and peer review*, 6 (2021), 1–8 (Springer).

Allaire, JJ, Yihui Xie, Christophe Dervieux, Jonathan McPherson, Javier Luraschi, Kevin Ushey, Aron Atkins, Hadley Wickham, Joe Cheng, Winston Chang, and Richard Iannone, "rmarkdown: Dynamic documents for r," (2025).

Brunson, Jason Cory, "ggalluvial: Layered grammar for alluvial plots," *Journal of Open Source Software*, 5 (2020), 2017.

Brunson, Jason Cory, and Quentin D. Read, "ggalluvial: Alluvial plots in 'ggplot2'," 2023.

Clancy, Matt, "The returns to science in the presence of technological risk," *arXiv preprint arXiv:2312.14289*, (2023).

Dycke, Nils, Ilia Kuznetsov, and Iryna Gurevych, "NLPeer: A unified resource for the computational study of peer review," in *Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: Long papers)*, Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, eds., 2023.

Eger, Steffen, Yong Cao, Jennifer D'Souza, Andreas Geiger, Christian Greisinger, Stephanie Gross, Yufang Hou, Brigitte Krenn, Anne Lauscher, Yizhi Li, Chenghua Lin, Nafise Sadat Moosavi, Wei Zhao, and Tristan Miller, "Transforming science with large language models: A survey on AI-assisted scientific discovery, experimentation, content generation, and evaluation," *arXiv preprint arXiv:2505.05151*, (2025).

Firke, Sam, "janitor: Simple tools for examining and cleaning dirty data," (2024).

Gagolewski, Marek, "stringi: Fast and portable character string processing in R," *Journal of Statistical Software*, 103 (2022), 1–59.

Gamer, Matthias, Jim Lemon, and Ian Fellows Puspendra Singh <puspendra.pusp22@gmail.com>, "irr: Various coefficients of interrater reliability and agreement," (2019).

Garnier, Simon, Ross, Noam, Rudis, Robert, Camargo, Antônio Pedro, Sciaini, Marco, Scherer, and Cédric, "viridis(Lite) - colorblind-friendly color maps for r," (2024).

Hester, Jim, and Jennifer Bryan, "glue: Interpreted string literals," (2024).

Korinek, Anton, "AI agents for economic research," *SSRN Working Paper*, (2025).

Luo, Ziming, Zonglin Yang, Zexin Xu, Wei Yang, and Xinya Du, "LLM4SR: A survey on large language models for scientific research," *arXiv preprint arXiv:2501.04306*, (2025).

Mitchener, Ludovico, Ethan Yiu, and others, "Kosmos: An AI scientist for autonomous discovery," *arXiv preprint arXiv:2511.02824*, (2025).

Müller, Kirill, "here: A simpler way to find your files," (2025).

Pataranutaporn, Pat, Nattavudh Powdthavee, Chayapatr Achiwaranguprok, and Pattie Maes, "Can AI solve the peer review crisis? A large scale cross model experiment of LLMs' performance and biases in evaluating over 1000 economics papers," 2025.

Pedersen, Thomas Lin, "ggforce: Accelerating 'ggplot2'," (2025a).

——, "patchwork: The composer of plots," (2025b).

R Core Team, "R: A language and environment for statistical computing," (Vienna, Austria, R Foundation for Statistical Computing, 2025).

Shuangbin Xu, Meijun Chen, Tingze Feng, Li Zhan, Lang Zhou, and Guangchuang Yu, "Use ggbreak to effectively utilize plotting space to deal with large datasets and outliers." *Frontiers in Genetics*, 12 (2021), 774846.

Sievert, Carson, "Interactive web-based data visualization with r, plotly, and shiny," (Chapman; Hall/CRC, 2020).

Slowikowski, Kamil, "ggrepel: Automatically position non-overlapping text labels with 'ggplot2'," (2024).

Son, Jaeho, and others, "FLAWS: A benchmark for error identification and localization in scientific papers," *arXiv preprint arXiv:2511.21843*, (2025).

Ushey, Kevin, JJ Allaire, and Yuan Tang, "reticulate: Interface to 'Python'," (2025).

Ushey, Kevin, and Hadley Wickham, "renv: Project environments," (2023).

Wickham, Hadley, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D'Agostino McGowan, Romain François, Garrett Grolemund, Alex Hayes, Lionel Henry, Jim Hester, Max Kuhn, Thomas Lin Pedersen, Evan Miller, Stephan Milton Bache, Kirill Müller, Jeroen Ooms, David Robinson, Dana Paige Seidel, Vitalie Spinu, Kohske Takahashi, Davis Vaughan, Claus Wilke, Kara Woo, and Hiroaki Yutani, "Welcome to the tidyverse," *Journal of Open Source Software*, 4 (2019), 1686.

Wickham, Hadley, Thomas Lin Pedersen, and Dana Seidel, "scales: Scale functions for visualization," (2025).

Xie, Yihui, "knitr: A comprehensive tool for reproducible research in R," in *Implementing reproducible computational research*, Victoria Stodden, Friedrich Leisch, and Roger D. Peng, eds. (Chapman; Hall/CRC, 2014).

——, "Dynamic documents with R and knitr," (Boca Raton, Florida, Chapman; Hall/CRC, 2015).

——, "knitr: A general-purpose package for dynamic report generation in R," (2025).

Xie, Yihui, J. J. Allaire, and Garrett Grolemund, "R markdown: The definitive guide," (Boca Raton, Florida, Chapman; Hall/CRC, 2018).

Xie, Yihui, Joe Cheng, Xianying Tan, and Garrick Aden-Buie, "DT: A wrapper of the JavaScript library 'DataTables'," (2025).

Xie, Yihui, Christophe Dervieux, and Emily Riederer, "R markdown cookbook," (Boca Raton, Florida, Chapman; Hall/CRC, 2020).

Zhang, Tianmai M, and Neil F Abernethy, "Reviewing scientific papers for critical problems with reasoning LLMs: Baseline approaches and automatic evaluation," *arXiv preprint arXiv:2505.23824*, (2025).

Zhang, Yaohui, Haijing Zhang, Wenlong Ji, Tianyu Hua, Nick Haber, Hancheng Cao, and Weixin Liang, "From replication to redesign: Exploring pairwise comparisons for LLM-based peer review," *arXiv preprint arXiv:2506.11343*, (2025).

Zhu, Hao, "kableExtra: Construct complex table with 'kable' and pipe syntax," (2024).

Data and methods

We draw on two sources: human evaluations from The Unjournal's public evaluation data, and LLM-generated evaluations produced by prompting frontier models with The Unjournal's rubric and requiring structured JSON output. We compare six models spanning three providers: GPT-5 Pro, GPT-5.2 Pro, and GPT-4o-mini (OpenAI); Claude Sonnet 4 and Claude Opus 4.6 (Anthropic); and Gemini 2.0 Flash (Google). Each model returns a midpoint rating and 90% credible interval for every metric, enforced by a strict JSON Schema.

Unjournal.org evaluations

We use The Unjournal's public data for a baseline comparison. At The Unjournal each paper is typically evaluated (aka 'reviewed') by two expert evaluators[2] who provide quantitative ratings on a 0–100 percentile scale for each of seven criteria (with 90% credible intervals),[3] two "journal tier" ratings on a 0.0 - 5.0 scale,[4] a written evaluation (resembling a referee report for a journal), and identification and assessment of the paper's "main claim". For our initial analysis, we extracted these human ratings and aggregated them, taking the average score per criterion across evaluators (and noting the range of individual scores).

All papers have completed The Unjournal's evaluation process (meaning the authors received a full evaluation on the Unjournal platform, which has been publicly posted at unjournal.pubpub.org). The sample includes papers spanning 2017–2025 working papers in development economics, growth, health policy, environmental economics, and related fields that The Unjournal identified as high-impact. Each of these papers has quantitative scores from at least one human evaluator, and many have multiple (2-3) human ratings.

LLM-based evaluation

Following The Unjournal's standard guidelines for evaluators and their academic evaluation form, evaluators are asked to consider each paper along the following dimensions: **claims & evidence**, **methods**, **logic & communication**, **open science**, **global relevance**, and an **overall** assessment. Ratings are interpreted as percentiles relative to serious recent work in the same area. For each metric, evaluators are asked for the midpoint of their beliefs and their 90% credible interval, to communicate their uncertainty. For the journal rankings measure, we ask both "what journal ranking tier should this work be published in? (0.0-5.0)" and "what journal ranking tier will this work be published in? (0.0-5.0)", with some further explanation.The full prompt can be seen in the code below – essentially copied from the Unjournal's guidelines page.

We captured the versions of each paper that was evaluated by The Unjournal's human evaluators, downloading from the links provided in The Unjournal's Coda database.

We evaluate each paper by passing the PDF directly to the model and requiring a strict, machine-readable JSON output. This keeps the assessment tied to the document the authors wrote. Direct ingestion preserves tables, figures, equations, and sectioning, which ad-hoc text scraping can mangle. It also avoids silent trimming or segmentation choices that would bias what the model sees.

[2] Occasionally they use 1 or 3 evaluators.

[3] See their guidelines here; these criteria include "Overall assessment", "Claims, strength and characterization of evidence", "Methods: Justification, reasonableness, validity, robustness", "Advancing knowledge and practice", "Logic and communication", "Open, collaborative, replicable science", and "Relevance to global priorities, usefulness for practitioners"

[4] "a normative judgment about 'how well the research should publish' " and "a prediction about where the research will be published"

JSON Schema

We enforce a strict JSON Schema for the output. The model must return one object for each metric, including a midpoint rating and a 90% credible interval. This guarantees that every paper is scored on the same fields with the same types and bounds. We request credible intervals (as we do for human evaluators) to allow the model to communicate its uncertainty rather than suggest false precision.

System Prompt

The system prompt is built from modular sections, each addressing a specific aspect of the evaluation task. We present each section below with brief explanations.

We instruct the model to act as an expert evaluator while explicitly preventing it from using author reputation, publication venue, or any prior knowledge of the paper's reception as evidence of quality.

Before scoring, we ask for a written assessment identifying key issues in the manuscript. This "think first, score second" approach encourages the model to ground its ratings in specific observations.

We define what percentile rankings mean and establish the reference group: "serious research in the same area encountered in the last three years." This anchors the model's judgments to a consistent baseline.

We define each of the seven percentile metrics, closely following The Unjournal's guidelines for evaluators. Note the emphasis on global priorities and practical relevance over pure academic novelty.

We explain what credible intervals mean and how to construct them. This guidance helps ensure the model produces well-calibrated uncertainty estimates rather than artificially narrow intervals.

In addition to percentile metrics, we ask for journal tier predictions on a 0-5 scale. These provide an external benchmark and, for papers that eventually publish, allow us to assess predictive accuracy.

Finally, we include instructions for self-consistency checks and the required JSON output format. The validation section encourages the model to ensure its scores align with its written assessment.

The sections above are concatenated to form the complete system prompt:

The `evaluate_paper` function uploads a PDF to the API and submits a background job for evaluation:

Relying on GPT-5 Pro, we use a single-step call with a reasoning model that supports file input. One step avoids hand-offs and summary loss from a separate "ingestion" stage. The model reads the whole PDF and produces the JSON defined above. We do not retrieve external sources or cross-paper material for these scores; the evaluation is anchored in the manuscript itself.

The Python pipeline uploads each PDF once and caches the returned file id keyed by path, size, and modification time. We submit one background job per PDF to the OpenAI Responses API with "high" reasoning effort and server-side JSON-Schema enforcement. Submissions record the response id, model id, file id, status, and timestamps.

We then polls job status and, for each completed job, retrieve the raw JSON object, and write the responses to disk.

Multi-Model Evaluation

To compare model performance, we collect evaluations from multiple providers. This allows us to assess whether different models exhibit systematic biases or calibration differences.

For OpenAI models without reasoning (e.g., GPT-4o), we use a simpler synchronous call:

Anthropic's API differs from OpenAI: PDFs are passed as base64-encoded content, and calls are synchronous (no background jobs). We use Claude's native PDF support.

Google's Gemini API supports PDF via file upload:

A unified runner collects evaluations from all providers:

GPT-5.2 Pro Focal Run (with Key Issues)

This section evaluates 14 focal papers using GPT-5.2 Pro with an extended schema that outputs a numbered list of key issues alongside the standard metrics.

Key Issues Comparison with Human Critiques

To validate how well the LLM identifies substantive issues, we compare its `key_issues` output against human expert critiques from The Unjournal's Coda database. Human expert critiques provide a high-quality reference standard for issue identification: they are produced by paid domain experts and synthesized by evaluation managers, but remain noisy and incomplete, so we treat them as a comparative signal rather than ground truth.

The comparison proceeds in two stages: first, we extract and align the data sources; then, optionally, we use an LLM to systematically assess the degree of alignment between machine-generated and human-identified issues.

The comparison pipeline uses a manually curated input and produces structured outputs:

Input: - **`results/key_issues_comparison.md`**: A manually curated markdown document pairing GPT-identified issues with human expert critiques for each paper.

Outputs: - **`results/key_issues_comparison.json`**: Parsed data in JSON format for programmatic use. - **`results/key_issues_comparison_results.json`**: LLM-assessed alignment metrics including coverage, precision, missed issues, and overall ratings.

The coverage metric estimates what fraction of human-identified issues the LLM captured; precision estimates what fraction of LLM issues are genuinely relevant. Together with the qualitative ratings, these provide a structured assessment of how well the LLM's issue identification aligns with expert judgment.

Results: Ratings

We compare quantitative ratings from 6 frontier LLMs (GPT-5 Pro, GPT-5.2 Pro, GPT-4o-mini, Claude Sonnet 4, Claude Opus 4.6, Gemini 2.0 Flash) against human expert reviews from The Unjournal. 45 papers have both human and LLM evaluations, enabling direct comparison across all seven percentile criteria and two journal-tier predictions.

We submit each PDF through a single, schema-enforced API call. The system prompt mirrors The Unjournal rubric, blocks extrinsic priors (authors, venue, web context), and requires strict JSON: a

~1k-word diagnostic summary plus numeric midpoints and 90% credible intervals for every metric.[5] Each numeric field carries a midpoint (model's 50% belief) and a 90% credible interval; wide intervals indicate acknowledged uncertainty.

Cost and token usage. Token costs are computed from API-reported input/output tokens (GPT-5 Pro runs with `reasoning` = high).

Table 1: Token usage and estimated cost per model

Model	Papers	Avg input	Avg output	Avg reasoning	Total cost	Cost/paper
Claude Opus 4.6	45	116,930	1,509	—	\$84.02	\$1.867
GPT-5 Pro	60	38,516	6,400	4,717	\$57.71	\$0.962
GPT-5.2 Pro	14	45,143	2,889	707	\$11.91	\$0.850
Claude Sonnet 4	16	113,798	727	—	\$5.64	\$0.352
GPT-4o-mini	60	71,581	470	—	\$0.66	\$0.011
Gemini 2.0 Flash	60	18,084	1,024	—	\$0.10	\$0.002

Total cost across all 255 LLM evaluations: **\$160.03**. Reasoning-capable models (GPT-5 Pro, GPT-5.2 Pro, Claude Opus 4.6) cost substantially more per paper than their lighter counterparts (GPT-4o-mini, Gemini 2.0 Flash), raising practical questions about cost-quality trade-offs for deployment at scale.

Overall ratings. Each point in Figure 4 represents a paper, with human mean rating on the x-axis and LLM rating on the y-axis. Points on the diagonal indicate perfect agreement. Human means carry their own variance; correlations are bounded by human inter-rater noise.

Most models cluster around the identity line in the 40–80 range, but diverge more at the extremes—LLMs tend to compress ratings toward the centre of the scale relative to humans, producing smaller spread. Outlier papers (those far from the diagonal) merit qualitative investigation; we flag the largest disagreements in the supplementary material.

Agreement metrics. We assess agreement using Pearson r (linear correlation), Spearman (rank-based, robust to outliers), mean bias (LLM $-$ Human; positive = LLM rates higher), and RMSE/MAE (error magnitude).

Table 2: Agreement between each LLM and human mean ratings (overall criterion)

Model	N	Pearson r	Spearman	Mean bias	RMSE	MAE
Claude Opus 4.6	33	0.567	0.477	-7.6	17.2	12.4
GPT-5 Pro	45	0.342	0.517	+6.4	15.4	10.3
Gemini 2.0 Flash	45	0.342	0.313	-5.3	14.5	11.7
GPT-5.2 Pro	9	0.333	0.471	-2.7	19.0	14.7
GPT-4o-mini	45	0.135	0.116	+0.0	20.2	13.1

[5] 0–100 percentiles relative to a reference group of ~'all serious work you have read in this area in the last three years', overall (global quality/impact), claims_evidence (claim clarity/support), methods (design/identification/robustness), advancing_knowledge (contribution to field/practice), logic_communication (internal consistency and exposition), open_science (reproducibility, data/code availability), and global_relevance (decision and priority usefulness). Journal tiers: (0–5, continuous) capture where the work should publish vs. will publish.

Table 2: Agreement between each LLM and human mean ratings (overall criterion)

Model	N	Pearson r	Spearman	Mean bias	RMSE	MAE
Claude Sonnet 4	11	0.020	0.055	+0.2	9.5	6.9

Table 2 reports correlation and error metrics between each LLM and the human mean overall rating. Positive mean bias indicates the model rates papers higher than humans on average; negative bias indicates harsher ratings. These correlations are bounded above by human inter-rater noise—since the human "mean" is itself noisy, no model can achieve a perfect correlation.

Human baseline context. To interpret human-LLM agreement, we compare against human-human agreement as an upper bound. If human evaluators only agree with each other at $= 0.5$, expecting LLMs to exceed that would be unrealistic. Table 3 shows $_{\text{HH}}$ (Krippendorff's alpha among human evaluators) alongside $_{\text{HL}}$ (between the human mean and each LLM).

Table 3: Human-human vs Human-LLM agreement by criterion (Krippendorff's)

Criterion	_HH	_HL (GPT-5 Pro)	_HL (GPT-5.2 Pro)	_HL (GPT-4o-mini)	_HL (Claude Sonnet 4)	_HL (Claude Opus 4.6)	_HL (Gemini 2.0 Flash)
Overall	0.55	0.25	0.36	0.14	0.06	0.47	0.25
Claims	0.42	0.20	0.26	0.19	0.21	0.26	0.18
Methods	0.54	0.29	0.07	0.03	-0.22	0.39	0.16
Adv. Knowledge	0.20	0.18	0.20	0.13	0.06	0.32	0.07
Logic & Comms	0.33	-0.04	-0.19	-0.00	0.26	0.09	-0.03
Open Science	0.03	-0.06	-0.61	-0.16	0.25	-0.18	0.03
Global Relevance	0.33	0.25	0.51	0.04	0.06	0.18	0.17

Where $_{\text{HL}}$ approaches $_{\text{HH}}$, the LLM is performing at or near the ceiling set by human inter-rater variability. Criteria where $_{\text{HH}}$ is already low (indicating genuine evaluator disagreement) naturally cap how well any model can do. Extended agreement analysis—including CI coverage, bootstrap confidence intervals, mutual information, and per-criterion correlations—is available in the supplementary material.

Criteria-level patterns. Figure 5 displays the mean rating each evaluator type assigns across the seven criteria. Criteria where all models and humans converge (similar colour intensity) suggest robust agreement on that dimension; criteria with divergent columns indicate systematic differences in how humans and LLMs weigh that aspect of quality.

Systematic rating differences. Figure 6 shows Human − LLM differences for each paper and criterion (primary model: GPT-5 Pro). Green indicates humans rated higher; orange indicates the LLM rated higher. Papers are ordered by their overall rating difference.

Figure 4: Human vs LLM overall ratings

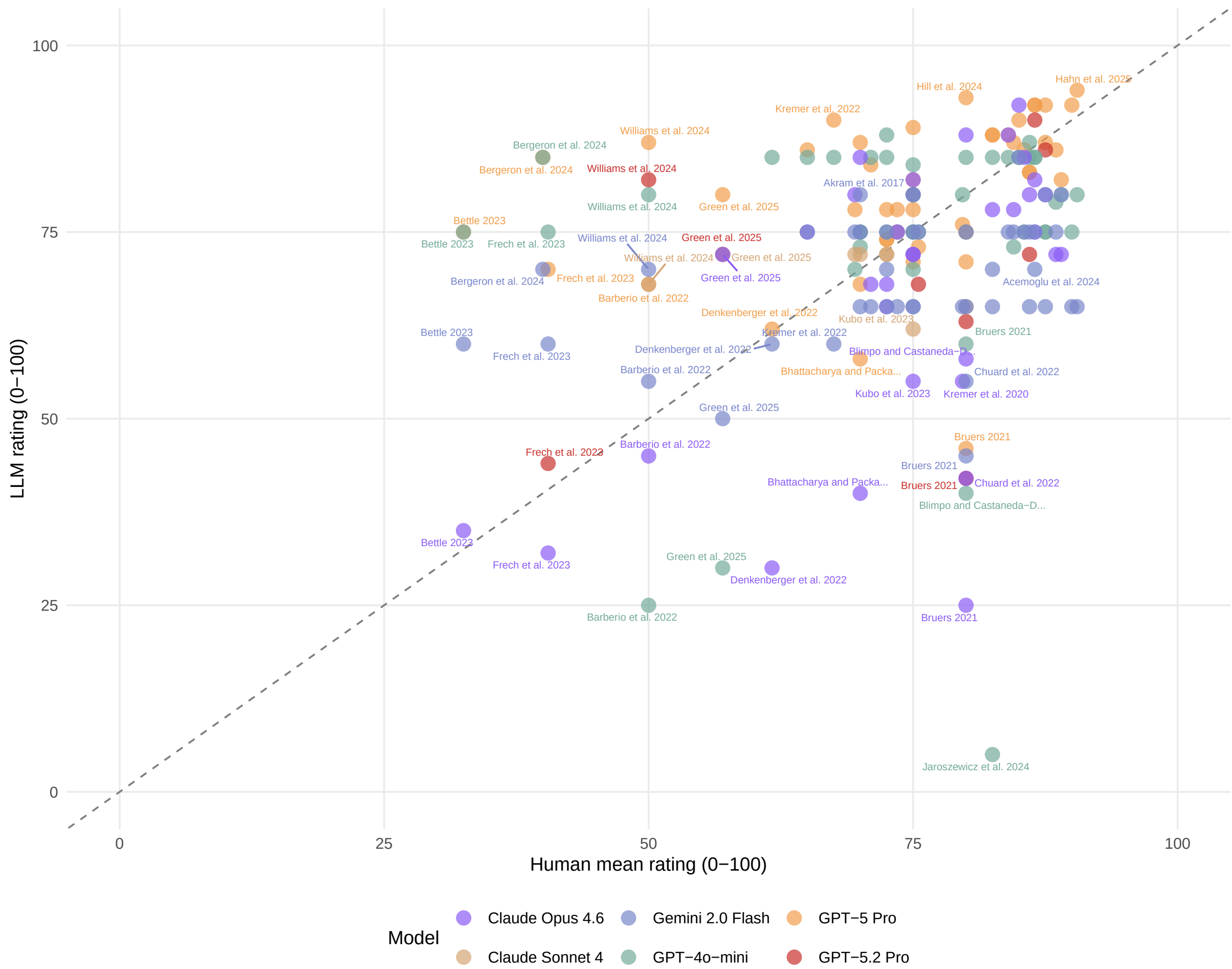


Figure 5: Mean rating by model and criterion

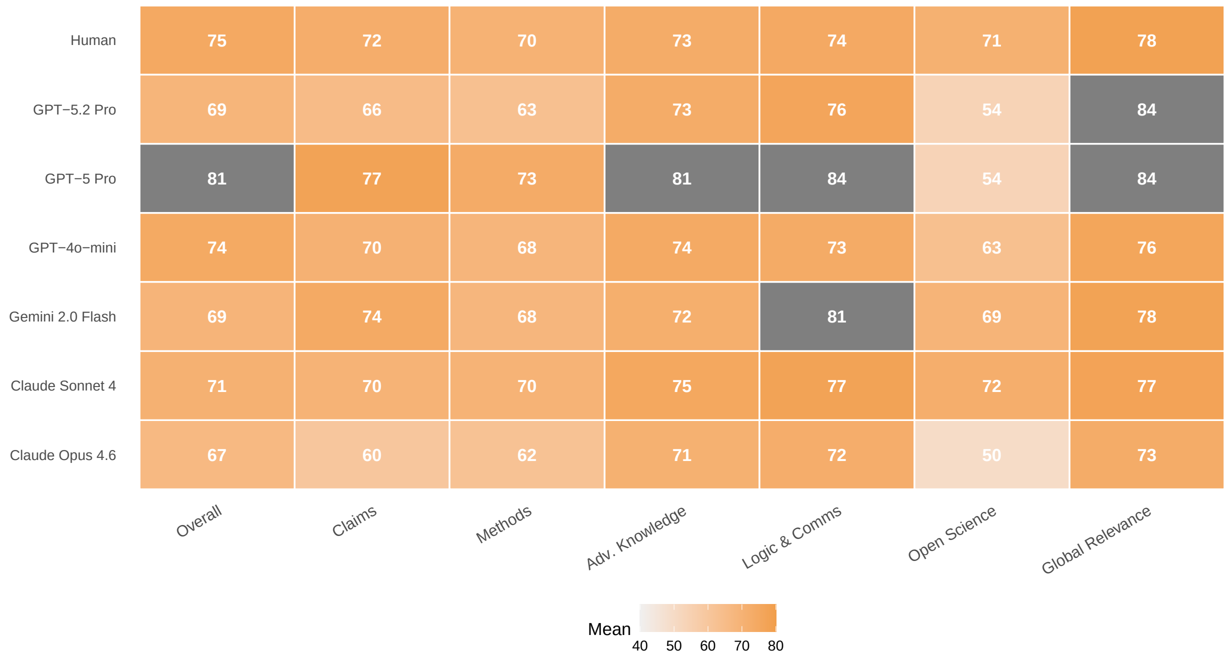
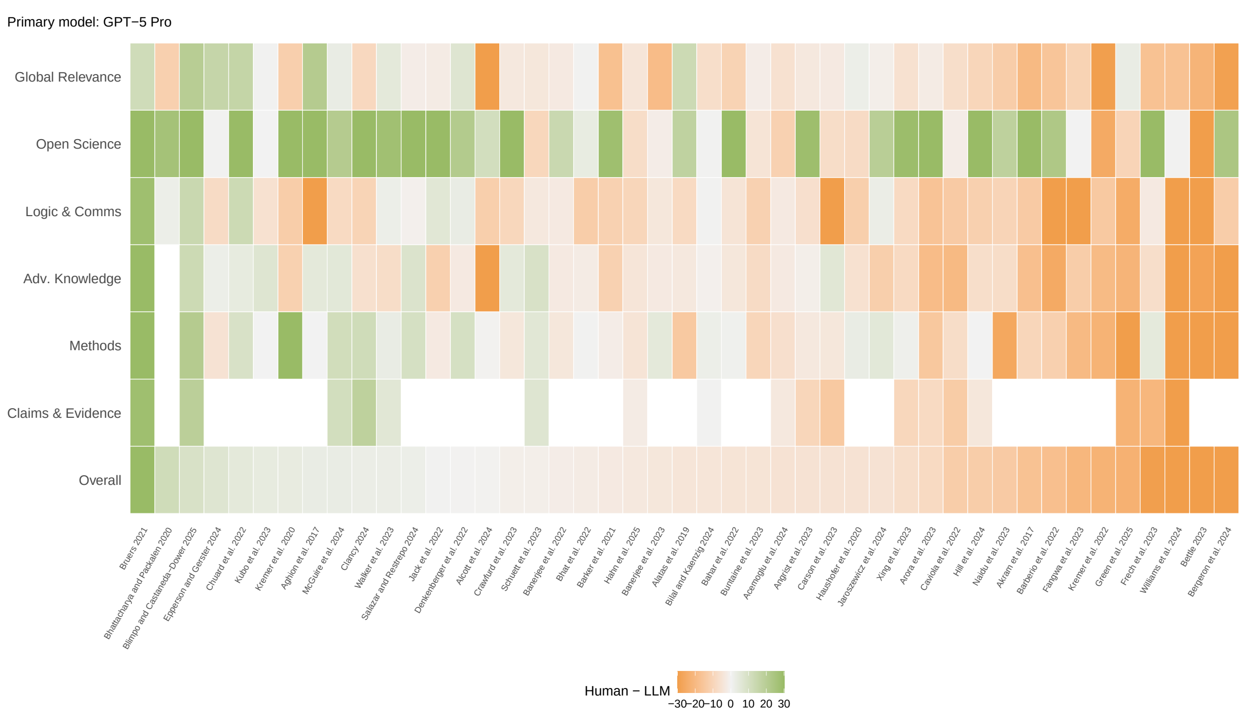


Figure 6: Human − LLM rating differences by paper and criterion



Rows with uniformly green or orange tiles indicate papers where humans and the LLM disagree systematically across all criteria, not just on one dimension. Columns with consistent colour suggest criteria-level biases (e.g., if the LLM consistently rates "open science" higher than humans).

GPT-5 Pro evaluations include extended reasoning traces. See the Appendix for full assessment summaries, reasoning traces, and extended statistical analysis (Krippendorff's alpha, bootstrap CIs, mutual information, tier correlations, top disagreement cases, and prompt comparison).

Results: Critiques & Key Issues

This chapter reports preliminary results from comparing LLM-identified key issues against human expert critiques. The analysis uses LLM-based assessment (GPT-5.2 Pro as judge) to measure coverage and precision; human validation of these alignment scores is underway.

> ⚠ Warning
>
> This section is under active development. Coverage and precision metrics below are LLM-assessed; manual annotation is in progress.

Data sources. We compare GPT-5.2 Pro key issues (structured output from the focal evaluation run, January 2026) against human expert critiques curated from The Unjournal's internal tracking database, where evaluation managers synthesize evaluator feedback using severity labels ("Necessary", "Optional but important", "Unsure").

We matched **14 papers** with both GPT-5.2 Pro key issues and human expert critiques.

LLM-based assessment. The comparison was assessed using GPT-5.2 Pro, which evaluated coverage (what proportion of human concerns GPT identified) and precision (whether GPT issues are substantive).

- **Coverage**: Proportion of *consensus* human issues that have any LLM match (match quality >= 30%)
- **Weighted Coverage**: Mean match quality across all human issues (treating unmatched issues as 0%)
- **Precision**: Proportion of LLM issues that match a human concern

> ℹ Caveat on Precision
>
> "Precision" only reflects whether LLM issues match the *consensus* human issues curated in Coda—those prioritized by evaluation managers as the most important concerns. Many LLM-identified issues may have been noted by one or more individual human evaluators but were not included in this curated set. A low precision score does not necessarily mean the LLM raised irrelevant concerns; it may simply reflect issues that weren't prioritized in the consensus summary.

> ℹ Interpretation Guide
>
> - **Coverage**: Percentage of human-identified issues that GPT also captured (in some form). Higher = GPT missed fewer human concerns.
> - **Precision**: Percentage of GPT issues that are substantive rather than generic or spuri-

ous. Higher = GPT's critiques are more targeted.
- **Overall Rating**: Qualitative assessment (Excellent/Good/Moderate/Poor) based on both coverage and precision metrics.

The coverage-precision figure above shows substantial variation across papers: some papers achieve high coverage and precision, while others show the LLM missing key human concerns or raising issues not in the expert consensus. Paper-by-paper comparisons with matched issue pairs, structural difference tables, and the manual annotation tool are available in the full online version.

Supplementary Material

Related Work

AI practical research applications

Luo et al. (2025) survey LLM roles from idea generation to peer review, including experiment planning and automated scientific writing. They highlight opportunities (productivity, coverage of long documents) alongside governance needs (provenance, detection of LLM-generated content, standardizing tooling) and call for reliable evaluation frameworks.

Korinek (2025) considers the specific potential for using AI in economic research, offering "working examples and step-by-step code" to help economists use agents for literature reviews, econometric code, fetching and analyzing economic data, and coordinate complex research workflows. He advocates economists using AI tools to boost productivity and "to understand [these tools'] capabilities and limitations." He calls for human researcher oversight "to ensure [these tools] pursue the right objectives and maintain research integrity". He outlines a range of current limitations and open questions (outlined here), including hallucinations, mistakes compounding through multi-agent workflows and "brittleness to small variations in prompts". At the higher level, he argues "the questions that will matter most—what problems deserve investigation, how should we evaluate trade-offs, what constitutes progress—will remain irreducibly human."

Mitchener, Yiu, et al. (2025) present Kosmos, "an AI scientist for autonomous discovery". However, evaluations reveal important limitations: Kosmos was found to be only 57% accurate in statements that required interpretation of results, and "identifying valuable discoveries Kosmos made is a time-intensive process that relies on human scientists with significant domain expertise."

AI peer review, error detection, and research evaluation

Son et al. (2025) evaluates LLMs on 83 papers paired with 91 confirmed significant scientific errors. Their strongest (OpenAI o3) model identified only 21.1% of these errors, with only 6.1% precision—domain experts found most false positives were hallucinations or student-level misconceptions.

Eger et al. (2025) provide a broad review of LLMs in science and a focused discussion of AI-assisted peer review. They argue: (i) peer-review data is scarce and concentrated in CS/OpenReview venues; (ii) targeted assistance that preserves human autonomy is preferable to end-to-end reviewing; and (iii) ethics and governance (bias, provenance, detection of AI-generated text) are first-class constraints.

Dycke, Kuznetsov, and Gurevych (2023) present NLPEER, a "multidomain corpus" of more than 5000 papers and 11k review reports from 2012 to 2022 from five different venues.

Zhang and Abernethy (2025) propose deploying LLMs as quality checkers to surface critical problems instead of generating full narrative reviews. Using papers from WITHDRARXIV and an automatic evaluation framework that leverages "LLM-as-judge," they find the best performance from top reasoning models but still recommend human oversight.

Pataranutaporn et al. (2025) asked four nearly state-of-the-art LLM models to consider 1220 unique papers drawn from 110 economics journals excluded from the training data of current LLMs. They find general alignment between LLM ratings and RePEC journal rankings, mixed results on detecting AI-generated papers, and evidence that LLMs show biases towards named high-prestige male authors and elite institutions.

Zhang et al. (2025) use AI conference paper data and LLM agents for pairwise manuscript comparisons, finding this approach outperforms traditional rating-based methods for identifying high-impact papers by citation metrics, though with some evidence of biases favoring established research institutions.

AI research taste and potential to guide research

A key motivation for this work stems from broader claims about AI's potential to accelerate scientific discovery. If AI systems are to contribute meaningfully to research—not just as tools but as evaluators and potential collaborators—we need to understand their "taste": what they value, what they miss, and whether their priorities align with human expert judgment and real-world research impact. This connects to broader debates about the returns to science in the presence of technological risk (Clancy 2023).

Extended Rating Analysis

The main text reports six core exhibits: cost overview, human-vs-LLM scatter plot, agreement metrics, human baseline (Krippendorff's alpha), criteria-level heatmap, and gap heatmap. The following extended analyses are preserved in the project's git history and available in earlier rendered versions:

- **Krippendorff's alpha by criterion**: Inter-rater agreement treating human mean and each LLM as separate raters
- **CI coverage analysis**: Share of human mean ratings falling within LLM 90% credible intervals
- **Bootstrap 95% CIs**: Nonparametric confidence intervals for Pearson correlations
- **Mutual information**: Information-theoretic measure of human-LLM rating dependence
- **Per-criterion correlations**: Pearson and Spearman by evaluation criterion and model
- **Tier correlation dumbbell plot**: How criteria predict journal tier judgments for humans vs. LLMs
- **Top disagreement cases**: Papers with largest human-LLM rating differences
- **Prompt comparison**: Legacy vs. updated GPT-5 Pro prompt effect on ratings

Extended Critique Analysis

The main text reports coverage-precision scatter and summary statistics. The following extended analyses are available in the full online version:

- **Paper-by-paper issue comparisons**: Interactive collapsible panels showing matched issue pairs, unmatched human/LLM issues, and detailed discussion for each paper
- **Aggregate statistics**: Total issue counts by severity, human vs. LLM issue distributions

- **Observable structural differences**: Format, structure, attribution, and length comparisons
- **Manual annotation tool**: Browser-based UI for systematic concordance labeling

---

The per-paper LLM evaluation traces below are only available in the online HTML version.